

Practice Midterm and Review Guidelines

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Generic Rules:

1. We have five big problems, each with a series of subproblems, usually four or five. (The practice problems here might have more subproblems. This is because I am nice and I'm happy to provide you with more practice materials.)
2. Write your answers in English. You should always explain any statement you made, and always show some process of calculation (which may help you earn some partial credit). This is a mid-term after all, so please be more formal and careful in your answers. We will give partial credits for partial solutions.
3. The exam is open book (allowing one sheet of A4 paper, double-sided is fine). No calculator is allowed.
4. No plagiarism allowed.

Problem 1 (Basic Concept Problem). *The first problem will see if you know what linear algebra is doing at all. Typically I would try to test your understanding of these subjects: linear maps and matrices, column/row operations, matrix-vector multiplication, RREF.*

I would set up the problem by one of the following ways:

1. *I might simply give you a matrix and want you to find RREF.*
2. *I might describe a geometric process, maybe some rotation/reflection/shearing/projection. Then I ask you to find its matrix.*
3. *I might describe a real life situation, maybe some analogue of the chicken-rabbit problem. Then I ask you to find the matrix for the linear map involved.*
4. *I might describe a matrix in terms of its columns or rows. Say $A = [\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$ with some given relations on $\mathbf{u}, \mathbf{v}, \mathbf{w}$. Maybe I tell you $\mathbf{u}, \mathbf{v}$ are not parallel, and $\mathbf{w} = \mathbf{u} + \mathbf{v}$, and I ask you to find RREF.*
5. *I might give you a matrix with unknown entries, then give you the RREF of a matrix, and ask you to work backwards to figure out the matrix.*
6. *I might ask you to find row/column operations to achieve a specific result.*
7. *I might want you to solve some linear systems.*
8. *i might ask about ranks.*
9. *The list is not exhaustive. You get the idea.*

Here is a specific example problem for you to get a feel. (For this whole problem, I personally would spend the most time doing the first subproblem, and then spend almost no time on the rest as I can do them by sight.)

Suppose $\text{RREF}(A)$ is the following matrix:

$$\text{RREF}(A) = \begin{bmatrix} 0 & 1 & 2 & 0 & 3 & 0 & 4 \\ 0 & 0 & 0 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$$

And we know some partial information about A , but the star portion is unknown.

$$A = \begin{bmatrix} * & * & 2 & 3 & * & * & 10 \\ * & * & 4 & 1 & * & * & 10 \\ * & 3 & * & * & 10 & * & 20 \\ * & * & 8 & * & 10 & * & 0 \end{bmatrix}.$$

1. Find A .
2. Starting from $\text{RREF}(A)$, we can perform column operations such that all non-pivot entries will become zero. Find the unit triangular matrix C representing these column operations.
3. Calculate AC .
4. Find the rank of AC^8 and $A(C - I)^7$. (Obviously brute force would waste lots of time, and hence not necessarily a good strategy.)

Answer:

Some answers are long NOT because the problems are hard. On the contrary, it means there are many different methods to get to the same answer.

1. The nature of RREF is to describe the column relations of the underlying matrix. So from $\text{RREF}(A)$, we see that in the matrix A , the first column is $\mathbf{0}$. The third column must be twice the second column. The fifth column must be three times the second column plus the fourth column. So on so forth. These are enough information to work out A as the following:

$$A = \begin{bmatrix} 0 & 1 & 2 & 3 & 6 & 0 & 10 \\ 0 & 2 & 4 & 1 & 7 & 0 & 10 \\ 0 & 3 & 6 & 1 & 10 & 2 & 20 \\ 0 & 4 & 8 & -2 & 10 & -4 & 0 \end{bmatrix}.$$

2. We want to use the pivotal columns to kill all non-pivotal columns. There are several ways to do this.

You can do it one operation at a time and combine them (the combination process is super easy, for reasons similar to how we figure out L in the LU decomposition).

Another way is simply to write out C one column at a time. Each column of C will tell you how to com-

bine columns of $\text{RREF}(A)$ into the corresponding column of $\text{RREF}(A)C = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.$

Alternatively, you can write a block matrix $\begin{bmatrix} \text{RREF}(A) \\ I \end{bmatrix}$ and do column operations. When the upper block turns into $\text{RREF}(A)C$, the lower block would turn into C simultaneously.

Either way, here is the answer.

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -2 & 0 & -3 & 0 & -4 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 & 0 & -2 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

3. Direct calculation. (If you want to go further, note that AC simply means kill all non-pivotal columns of A . Then you can write out the answer with no calculation needed.)

$$AC = \begin{bmatrix} 0 & 1 & 0 & 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 1 & 0 & 2 & 0 \\ 0 & 4 & 0 & -2 & 0 & -4 & 0 \end{bmatrix}.$$

4. As a rule of thumb, when you see big numbers, it means there should be some tricks involved.

C is invertible so it will not change rank. So AC^8 has the same rank as A , i.e., 3. And you might be able to tell by sight that $(C - I)^7$ is the zero matrix (we had a daily problem about this on October 6th).

Problem 2 (Inverse Matrix Problem). *The second problem will focus on matrix multiplications, inverse matrices, LDU decompositions.*

I would set up the problem in one of the following ways:

1. *I might give you a matrix similar to one of the methods in problem 1.*
2. *I might give you several matrices, and ask you to find legal ways to multiply them.*
3. *I might ask you to find inverse matrices, find LDU decompositions, or show why something does not have an LDU decomposition. (In the last case, the matrix will be designed to be invertible, so we have a nice criteria here.)*
4. *I might work backwards: given you the inverse matrix and let you work out the original matrix. Or maybe tell you that an invertible matrix with partial information has no LDU, and you should work it out.*
5. *Sometimes an understanding of block matrices might help.*
6. *I might want you to find inverses in unorthodox ways. Say, if I start with $\begin{bmatrix} A \\ I \end{bmatrix}$, what would happen if I do column operations until the upper block is I ?*
7. *The list is not exhaustive. You get the idea.*

Here is a specific example problem for you to get a feel. (For this whole problem, I would probably spend most time on the LDU decomposition of A and finding its inverse. The rest is easy.)

We have a matrix A here, where a is some unknown real number.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ a & 4 & 5 \\ 2a & 8 & 6 \end{bmatrix}.$$

1. *When is A not invertible?*

2. When A is not invertible, find the RREF for the augmented matrix of the system $A\mathbf{x} = \begin{bmatrix} 5 \\ 9 \\ 14 \end{bmatrix}$.
3. When A is invertible, find its LDU decomposition, or show that LDU decomposition does not exist.
4. When A is invertible, calculate $\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} A^{-1}$.
5. Note that $B = \begin{bmatrix} 1 & -\frac{3}{4} & \frac{1}{8} \\ 0 & -\frac{3}{8} & \frac{5}{16} \\ 0 & \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$ is the inverse of A when $a = 0$. Let C be the inverse of A when $a = 1$. Find $C - B$.

Answer:

Some answers are long NOT because the problems are hard. On the contrary, it means there are many different methods to get to the same answer.

1. There are several methods. The brute force way is to do gaussian elimination. Then REF is the following

$$\begin{bmatrix} 1 & 2 & 3 \\ & 4 - 2a & 5 - 3a \\ & & -4 \end{bmatrix}.$$

So we have three pivots iff $4 - 2a \neq 0$, iff $a \neq 2$. So A is not invertible only when $a = 2$.

The second way is to stare, and observe that when $a = 2$, the first two columns are parallel.

The third way is to use determinant, but it sucks for three reasons: first, we have not learned about it yet. Second, it is slower. (It is essentially a slower but equivalent way to do gaussian elimination, i.e., the first method here.) Third, it indicates a lack of understanding of the nature of invertibility. Nevertheless, if you correctly find the determinants $8a - 16$, you will get all the credits for this subproblem.

2. You can directly do Gaussian elimination to $\begin{bmatrix} 1 & 2 & 3 & 5 \\ 2 & 4 & 5 & 9 \\ 4 & 8 & 6 & 14 \end{bmatrix}$ and find RREF to be $\begin{bmatrix} 1 & 2 & & 2 \\ & & 1 & 1 \\ & & & 0 \end{bmatrix}$.

You can also observe column relations of $\begin{bmatrix} 1 & 2 & 3 & 5 \\ 2 & 4 & 5 & 9 \\ 4 & 8 & 6 & 14 \end{bmatrix}$ to find RREF. Note that the last column is the sum of the second and third column. This is probably the fastest method for this specific subproblem.

If you did last subproblem by REF, then you also have a readily available LU decomposition to simplify the process. We have

$$A = LU = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ & 0 & -1 \\ & & -4 \end{bmatrix}.$$

Now the same operation turning A to U would turn $\begin{bmatrix} 5 \\ 9 \\ 14 \end{bmatrix}$ to $\begin{bmatrix} 5 \\ -1 \\ -4 \end{bmatrix}$. So we simply need to do RREF

of $\begin{bmatrix} 1 & 2 & 3 & 5 \\ & 0 & -1 & -1 \\ & & -4 & -4 \end{bmatrix}$, and get the same answer. (This will be slightly faster than the first method, because you essentially saved some calculation steps.)

3. (If you did not use REF for the first subproblem, well, you have to do it here anyway. I usually design problems that reward methods focusing on an in depth analysis of a specific matrix.) We already see from before that

$$A = LU = \begin{bmatrix} 1 & & \\ a & 1 & \\ 2a & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ & 4-2a & 5-3a \\ & & -4 \end{bmatrix} = \begin{bmatrix} 1 & & \\ a & 1 & \\ 2a & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ & 4-2a & 0 \\ & & -4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ & 1 & \frac{5-3a}{4-2a} \\ & & 1 \end{bmatrix}.$$

4. Direct calculation, but you can be slightly faster if you realize that the problem simply asks about the last row of A^{-1} . To calculate inverse, we can go from $\begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ a & 4 & 5 & 0 & 1 & 0 \\ 2a & 8 & 6 & 0 & 0 & 1 \end{bmatrix}$ to its RREF (and we also already know what the first two operations should be). Somewhere along the way, you shall see $\begin{bmatrix} 1 & 2 & 3 & 1 & 0 & 0 \\ & 4-2a & 5-3a & -a & 1 & 0 \\ & & 1 & & \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$, and you can already stop. No more operations would change the last row ever. So the last row of A^{-1} is always $[0 \quad \frac{1}{2} \quad -\frac{1}{4}]$.

You can of course keep going (but this would be unnecessary). The full RREF is

$$\begin{bmatrix} 1 & 0 & 0 & \frac{2}{2-a} & \frac{-3}{4-2a} & \frac{1}{8-4a} \\ 0 & 1 & 0 & \frac{-a}{4-2a} & \frac{3a-3}{8-4a} & \frac{5-3a}{16-8a} \\ 0 & 0 & 1 & 0 & \frac{1}{2} & -\frac{1}{4} \end{bmatrix}.$$

Finally, there is an alternative method. If the last row of A^{-1} is $\mathbf{x}^T$, then $\mathbf{x}$ should be perpendicular to the first two columns of A (this is because $A^{-1}A = I$). Taking a cross product of those two columns gives $\begin{bmatrix} 0 \\ 4a-8 \\ 4-2a \end{bmatrix}$, which is parallel to $\begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$. So the last row of A^{-1} must be $[0 \quad 2k \quad -k]$. Now this should also multiply the last column of A to 1, therefore $10k - 6k = 1$. So $k = \frac{1}{4}$ and the last row of A^{-1} must be $[0 \quad \frac{1}{2} \quad -\frac{1}{4}]$.

5. Two methods. You can calculate $C = \begin{bmatrix} 2 & -\frac{3}{2} & \frac{1}{4} \\ -\frac{1}{2} & 0 & \frac{1}{4} \\ 0 & \frac{1}{2} & -\frac{1}{4} \end{bmatrix}$ directly (especially if you get the formula for

$$A^{-1} \text{ in the last subproblem}), \text{ and simply subtract to get } C - B = \begin{bmatrix} 1 & -\frac{3}{4} & \frac{1}{8} \\ -\frac{1}{2} & \frac{3}{8} & -\frac{1}{16} \\ 0 & 0 & 0 \end{bmatrix}.$$

Or you can use Sherman-Morrison. We first observe that

$$A_1 = A_0 + \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}.$$

Therefore

$$C = B - \frac{1}{1 + \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} B \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}} B \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} B.$$

So

$$C - B = -2 \left(\begin{bmatrix} -\frac{1}{2} \\ \frac{1}{4} \\ 0 \end{bmatrix} \begin{bmatrix} 1 & -\frac{3}{4} & \frac{1}{8} \end{bmatrix} \right) = \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{bmatrix} \begin{bmatrix} 1 & -\frac{3}{4} & \frac{1}{8} \end{bmatrix}.$$

Problem 3 (Block Matrix Problem). *The third problem will focus on block matrices.*

I would set up the problem in one of the following ways:

1. *I might give you a matrix in block format, and ask various questions such as block row/column operations, block LDU, block matrix multiplication etc..*
2. *I might give you a big matrix, but you should spot the blocks yourself. Then the problems will be easier if you do block matrices stuff.*
3. *You need to be proficient in matrix multiplications. In particular, beware of the order of multiplications.*
4. *The list is not exhaustive. You get the idea.*

Here is a specific example problem for you to get a feel. (If you get the geometric intuition right, surprisingly little actual calculation is involved. But even without geometric intuitions, this is not too bad. I personally would probably waste some time on the last subproblem, trying various row operations to get it to work.)

Let $\mathbf{n}$ be a unit normal vector to some hyperplane through the origin in $\mathbb{R}^n$. Let $H = I - 2\mathbf{n}\mathbf{n}^T$ be the reflection about this hyperplane, and let $P = I - \mathbf{n}\mathbf{n}^T$ be the projection to this hyperplane.

We have matrices $A = \begin{bmatrix} H & \mathbf{n} \\ \mathbf{n}^T & 0 \end{bmatrix}$ and $B = \begin{bmatrix} P & \mathbf{n} \\ \mathbf{n}^T & 1 \end{bmatrix}$.

1. *Show that $HP = PH = P$ (you can calculate directly or give a geometric argument).*
2. *Show that $\mathbf{n}^T P = \mathbf{0}^T$ and $\mathbf{n}^T H = -\mathbf{n}^T$.*
3. *Calculate and simplify AB .*
4. *Show that B has no LU decomposition.*
5. *Find a block LDU decomposition of A .*
6. *Find the inverses for L, D, U in the LDU decomposition of A , and calculate $U^{-1}D^{-1}L^{-1}B^{-1}$.*
7. *With two elementary block row operations L_1, L_2 , I can have $L_2 L_1 A = B$. What are these two operations?*

Answer:

Some answers are long NOT because the problems are hard. On the contrary, it means there are many different methods to get to the same answer.

1. Geometrically, “reflect then project” is the same as “project directly”. Same for the other order. But of course, direct calculation always work:

$$HP = (I - 2\mathbf{n}\mathbf{n}^T)(I - \mathbf{n}\mathbf{n}^T) = I - 3\mathbf{n}\mathbf{n}^T + 2(\mathbf{n}\mathbf{n}^T)^2 = I - \mathbf{n}\mathbf{n}^T = P.$$

Same for the other order. Note that $(\mathbf{n}\mathbf{n}^T)^2 = \mathbf{n}\mathbf{n}^T$ because it is also a projection, or you can use the law of associativity $\mathbf{n}(\mathbf{n}^T \mathbf{n})\mathbf{n}^T = \mathbf{n}\mathbf{n}^T$.

2. Note that P is symmetric (because I and $\mathbf{n}\mathbf{n}^T$ are symmetric). Geometrically, we obviously have $P\mathbf{n} = \mathbf{0}$, and take transpose would give the result.

Alternatively, just do a direct calculation

$$\mathbf{n}^T P = \mathbf{n}^T (I - \mathbf{n}\mathbf{n}^T) = \mathbf{n}^T - \mathbf{n}^T \mathbf{n}\mathbf{n}^T = \mathbf{n}^T - (\mathbf{n}^T \mathbf{n})\mathbf{n}^T = \mathbf{0}^T.$$

The other part $\mathbf{n}^T H = -\mathbf{n}^T$ is done by the same methods.

3. The previous two subproblems are set-ups for this. (Which should not be a surprise....)

$$AB = \begin{bmatrix} HP + \mathbf{n}\mathbf{n}^T & H\mathbf{n} + \mathbf{n} \\ \mathbf{n}^T P & \mathbf{n}^T \mathbf{n} \end{bmatrix} = \begin{bmatrix} I & \\ & 1 \end{bmatrix}.$$

Looking back, why would I design these two block matrices if this is not the case? Hahaha.

4. B has no LU decomposition, because it is invertible, but its $(n-1)$ -th leading principal minor is not invertible (a projection).
5. How to do block elimination on A ? We need to use the upper left block H to eliminate the entry $\mathbf{n}^T$ below it. So we need to multiply H by $-\mathbf{n}^T H^{-1}$. Note that $H^{-1} = H$ and $\mathbf{n}^T H = -\mathbf{n}^T$. So along this line of thoughts, we eventually have

$$A = \begin{bmatrix} I & \\ -\mathbf{n}^T & 1 \end{bmatrix} \begin{bmatrix} H & \\ & 1 \end{bmatrix} \begin{bmatrix} I & -\mathbf{n} \\ & 1 \end{bmatrix}.$$

The minus sign is the easiest to get wrong. Make sure you double check your result by actually doing this multiplication and see if you indeed get A .

6. The inverses are $\begin{bmatrix} I & \\ \mathbf{n}^T & 1 \end{bmatrix}$, $\begin{bmatrix} H & \\ & 1 \end{bmatrix}$ and $\begin{bmatrix} I & \mathbf{n} \\ & 1 \end{bmatrix}$. And we have $U^{-1}D^{-1}L^{-1}B^{-1} = A^{-1}B^{-1} = (BA)^{-1} = I$.
7. This is a bit tricky, but it goes like this:

$$\begin{bmatrix} H & \mathbf{n} \\ \mathbf{n}^T & 0 \end{bmatrix} \rightarrow \begin{bmatrix} P & \mathbf{n} \\ \mathbf{n}^T & 0 \end{bmatrix} \rightarrow \begin{bmatrix} P & \mathbf{n} \\ \mathbf{n}^T & 1 \end{bmatrix}.$$

The two operations are $L_1 = \begin{bmatrix} I & \mathbf{n} \\ & 1 \end{bmatrix}$ and $L_2 = \begin{bmatrix} I & \\ \mathbf{n}^T & 1 \end{bmatrix}$. Note that $\mathbf{n}^T P = \mathbf{0}^T$.

Problem 4 (Subspace Problem). *The fourth problem will focus on the fundamental subspaces of matrices, i.e., $\text{Ran}(A)$, $\text{Ker}(A)$, $\text{Ran}(A^T)$, $\text{Ker}(A^T)$. However, this will be mixed with previous concepts such as ranks, basis, dimensions, row/column operations and such.*

Here is a specific example problem for you to get a feel. (I personally would fly through these questions super fast. Most time would be spent on doing REF for BA .)

We have a matrix

$$A = \begin{bmatrix} 3 & 3 & 3 & 3 \\ 3 & 2 & 2 & 2 \\ 3 & 2 & 1 & 1 \\ 3 & 2 & 1 & 0 \end{bmatrix}.$$

1. Find a matrix B such that BA is exactly the first three rows of A .
2. Find the subspaces $\text{Ran}(BA)$, $\text{Ker}(BA)$, $\text{Ran}((BA)^T)$, $\text{Ker}((BA)^T)$. (This means you should give me a basis for each subspace.)
3. Find a unit lower triangular matrix L and a unit upper triangular matrix U , such that $BA = LXU$ and X only has three non-zero entries.
4. Find the projection matrix to the kernel of BA . This should be a 4 by 4 matrix.

Answer:

Some answers are long NOT because the problems are hard. On the contrary, it means there are many different methods to get to the same answer.

1. $B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$. This is trivial.

2. $BA = \begin{bmatrix} 3 & 3 & 3 & 3 \\ 3 & 2 & 2 & 2 \\ 3 & 2 & 1 & 1 \end{bmatrix}$, and its RREF is $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$. Now, RREF and the original matrix have the same row space and same kernel, so $\text{Ran}((BA)^T)$ has basis $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3 + \mathbf{e}_4$, and $\text{Ker}(BA)$ has basis $\mathbf{e}_3 - \mathbf{e}_4$. To find column space and left kernel, note that BA is rank 3, so the column space is $\mathbb{R}^3$ and the left kernel is $\{\mathbf{0}\}$.

3. This is basically Gaussian elimination. $L = \begin{bmatrix} 1 & & & \\ 1 & 1 & & \\ 1 & 1 & 1 & \end{bmatrix}$, then $L^{-1}A = \begin{bmatrix} 3 & 3 & 3 & 3 \\ & -1 & -1 & -1 \\ & & -1 & -1 \end{bmatrix}$. Then we want column operations, so $U = \begin{bmatrix} 1 & 1 & 1 & 1 \\ & 1 & 1 & 1 \\ & & 1 & 1 \\ & & & 1 \end{bmatrix}$, and $L^{-1}AU^{-1} = \begin{bmatrix} 3 & 0 & 0 & 0 \\ & -1 & 0 & 0 \\ & & -1 & 0 \end{bmatrix}$ is the desired X .

Note that the method to get L and U is simply by reading the row/column operations that we are doing, i.e., how we find L in any LU decomposition. This should not take much time.

4. The kernel is the span of $\begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}$. The projection to the direction of a vector $\mathbf{w}$ has the matrix $\frac{\mathbf{w}\mathbf{w}^T}{\mathbf{w}^T\mathbf{w}}$.

So the answer is $\frac{1}{2} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$.

Problem 5 (Abstract Space Problem). *The fifth problem will focus on abstract spaces, basis and coordinates, linear maps and matrix representations, and all the change of coordinate processes.*

I usually prefer to use the following abstract spaces:

1. *Space of functions. Typically polynomial spaces, or the solution space to some differential equations. For linear maps, my favorites are derivative, or $f(x) \mapsto xf(x)$, or variants (linear combinations?) of them.*
2. *Space of matrices. Maybe upper triangular matrices. Maybe symmetric matrices. Etc. For linear maps, my favorites are $X \mapsto AX$ or XA for some fixed A , or transpose, or trace, or variants (linear combinations?) of them.*
3. $\mathbb{R}^n$ but under some exotic basis. Linear maps may have geometric meaning.
4. Maybe some applications on other sciences.
5. Maybe linear maps connecting the various spaces above.
6. The list is not exhaustive. You get the idea.

Here is a specific example problem for you to get a feel. (The last two subproblems can be time consuming. The second subproblem can be fast if you realize that the matrix times itself is $4I$.)

Let V be the space of all 2×2 matrices. We have a basis $\mathcal{B} = (E_{11}, E_{12}, E_{21}, E_{22})$ where $E_{ij} = \mathbf{e}_i \mathbf{e}_j^T \in V$. We also have a basis $\mathcal{C}$ made of the following matrices.

$$C_1 = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, C_2 = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, C_3 = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}, C_4 = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$. Then we have a linear map $L : V \rightarrow V$ such that $L(X) = AX - XA$. Let W be collection of all matrices X such that $AX = XA$. Then $W = \text{Ker}(L)$.

1. In order to verify that L is indeed a linear map, show that whenever $X, Y \in V$, then $L(X + Y) = L(X) + L(Y)$. And for any $k \in \mathbb{R}$ and any $X \in V$, we have $L(kX) = kL(X)$.
2. Find the change of coordinate matrix from basis $\mathcal{C}$ to basis $\mathcal{B}$, and from basis $\mathcal{B}$ to basis $\mathcal{C}$.
3. Find the matrix for the linear map L under basis $\mathcal{B}$, find its RREF, and then describe elements of W .
4. Find the matrix for the linear map L where the domain is under basis $\mathcal{C}$, and the codomain is under basis $\mathcal{B}$.

Answer:

Some answers are long NOT because the problems are hard. On the contrary, it means there are many different methods to get to the same answer.

1. Direct verification.

$$L(X+Y) = A(X+Y) - (X+Y)A = AX + AY - XA - YA = (AX - XA) + (AY - YA) = L(X) + L(Y).$$

$$L(kX) = A(kX) - (kX)A = kAX - kXA = k(AX - XA) = kL(X).$$

2. The most tricky part is to NOT confuse the two.

First let us find the change of coordinate matrix from basis $\mathcal{C}$ to basis $\mathcal{B}$. This means we are writing elements of $\mathcal{C}$ and express them in the coordinates under basis $\mathcal{B}$. This is the easy direction. So we have

$$I_{\mathcal{B} \leftarrow \mathcal{C}} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}.$$

Conversely, to find the change of coordinate matrix from basis $\mathcal{B}$ to basis $\mathcal{C}$, we should try to express elements of $\mathcal{B}$ in the coordinates under basis $\mathcal{C}$. So we have

$$E_{11} = \frac{1}{4}(C_1 + C_2 + C_3 + C_4).$$

$$E_{12} = \frac{1}{4}(C_1 - C_2 + C_3 - C_4).$$

$$E_{21} = \frac{1}{4}(C_1 + C_2 - C_3 - C_4).$$

$$E_{22} = \frac{1}{4}(C_1 - C_2 - C_3 + C_4).$$

So we have

$$I_{\mathcal{C} \leftarrow \mathcal{B}} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}.$$

Alternatively, we can use $I_{\mathcal{C} \leftarrow \mathcal{B}} = I_{\mathcal{B} \leftarrow \mathcal{C}}^{-1}$ to get the same answer.

3. Suppose $X \in V$ has coordinates $\begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}$ under basis $\mathcal{B}$. Then $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. So

$$L(X) = AX - XA = \begin{bmatrix} c-b & d-a-b \\ a+c-d & b-c \end{bmatrix}.$$

$$\text{So } L_{\mathcal{B} \leftarrow \mathcal{B}} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} c-b \\ d-a-b \\ a+c-d \\ b-c \end{bmatrix}. \text{ So}$$

$$L_{\mathcal{B} \leftarrow \mathcal{B}} = \begin{bmatrix} 0 & -1 & 1 & 0 \\ -1 & -1 & 0 & 1 \\ 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 0 \end{bmatrix}.$$

Gaussian elimination shows RREF as

$$\begin{bmatrix} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

So the kernel is made of elements of the form $\begin{bmatrix} y-x & x \\ x & y \end{bmatrix}$ for any $x, y \in \mathbb{R}$. Equivalently, a basis for W is made of $\begin{bmatrix} -1 & 1 \\ 1 & 0 \end{bmatrix}$ and I . (Another basis is simply A and I .)

4. Direct calculation

$$L_{\mathcal{B} \leftarrow \mathcal{C}} = L_{\mathcal{B} \leftarrow \mathcal{B}} I_{\mathcal{B} \leftarrow \mathcal{C}} = \begin{bmatrix} 0 & 2 & -2 & 0 \\ -1 & -1 & -3 & 1 \\ 1 & 3 & 1 & -1 \\ 0 & -2 & 2 & 0 \end{bmatrix}.$$

Alternatively, we can directly express $L(C_1), L(C_2), L(C_3), L(C_4)$ under basis $\mathcal{B}$. Then we have

$$\begin{aligned} L(C_1) &= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \\ L(C_2) &= \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix} \\ L(C_3) &= \begin{bmatrix} -2 & -3 \\ 1 & 2 \end{bmatrix} \\ L(C_4) &= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}. \end{aligned}$$

This gives the same four columns